Lodestar

Bankable demand forecasts that tell mini-grid developers and their lenders how much power a village will use, before anyone builds.

BRIEF
03 · Predict mini-grid demand before building

TRACK
2 · Digital Solutions & AI

TEAM
Tushar Agarwal · product, data and engineering

HOW THIS PITCH IS SCORED

Impact	25%
Innovation	25%
Sustainability	25%
Feasibility	15%
Scalability	10%

IMPACT · 25% OF SCORE

Villages get mini-grids sized to real demand

THE PROBLEM

490 million people need the 217,000 mini-grids still to be built (ESMAP). Each is sized before a single customer exists, so developers guess demand: oversize and tariffs price villagers out; undersize and they get evening brownouts.

"Survey-based estimates miss individual customers' demand by 65%." (CrossBoundary Innovation Lab)

5 of 6

site-years of real metered demand inside our range, forecast from open data with no site meters (DR Congo)

+37% → +6%

year-3 oversizing at a metered site: desk estimate vs Lodestar plan (72% less mismatch)

−48%

cost of power when sized on P50 instead of the optimistic case (avg. of 20 villages)

Bottom line: On real meters, demand landed inside our range in 5 of 6 site-years, and right-sizing roughly halves the cost of power.

INNOVATION · 25% OF SCORE

A P90 for demand, borrowed from solar finance

TODAY

Surveys or flat benchmarks give one day-one number: no range, no growth, never checked against meters. Lenders cannot price a guess.

OUR APPROACH

We forecast connections × consumption month by month with a calibrated P90/P50/P10 range, like a solar yield report, validated on real meters.

ROOT CAUSE WE ADDRESS

No defensible demand number before building, so sites are mis-sized or never financed.

HOW IT WORKS

1. Click any village: building footprints and OpenStreetMap give customers and anchors
2. 20,000 growth scenarios give P90/P50/P10 for 5 years, explained in plain words
3. Size the system, stress-test the loan; shared meters sharpen it (Demand Bureau)

Bottom line: A calibrated range, a growth curve and a lender view, validated on meters: none of the open tools we found do all four.

SUSTAINABILITY & AFFORDABILITY · 25% OF SCORE

Free to villages, pays for itself at scale

COST TO THE USER

$0 / month

for households. Developers pay $300 per site report: 0.05% of a ~$585k mini-grid.

PAYBACK

~200 reports/yr

WHO PAYS

First: developers bidding into results-based financing, per report. Next: RBF programmes, $12k/yr screening. Later: lenders.

HOW IT PAYS FOR ITSELF

Open data, 0.76 CPU-s per forecast. Break-even ~200 reports/yr: 3% of 6,116 planned sites (AMDA). No grant to run.

WHO KEEPS IT RUNNING

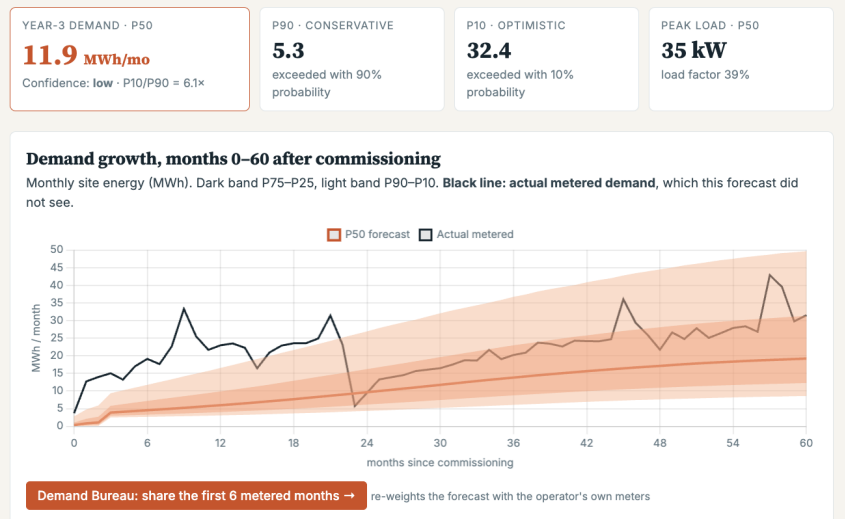
Two analysts in Nairobi or Lagos; Python on a laptop, no API keys; quarterly re-fit on Bureau meter data.

ENVIRONMENTAL BENEFIT

No stranded panels and batteries (desk plan oversized 61 kWp at one site, ours 9) and less backup diesel.

Bottom line: Free for villagers, under a second of compute per forecast, and self-funding at about 200 paid reports a year.

Live today, validated on real meters



EVIDENCE IT WORKS

- DRC meters: demand in range 5 of 6 site-years
- HOMER export; open licences; aggregate-only data
- Risk: 3 metered sites → backtest operators first

PATH TO SCALE

PILOT

Nigeria: 10 sites with developers bidding into DARES

EXPAND

Score on meters; one RBF programme; Kenya next

SCALE

5 countries via the Bureau; ~1 CPU-s per site

OUR ASK

2–3 mini-grid operators to share historical meter data, and one RBF programme as pilot partner. Live: lodestar-utfz.onrender.com

Bottom line: Live today on real data; the next step is a 10-site pilot with developers bidding into Nigeria's DARES programme.